Multi Objective Airfoil Optimizer using XFOIL

Overview

This is a script for performing multi objective airfoil optimization using XFOIL. Currently allows the user to optimize for maximum C_l and operational window, defined as the range of angle of attacks where $C_l \geq 0.9 * C_{l_{max}}$

The user has the option to provide a baseline airfoil, which must be provided in the PARSEC airfoil parametrization format. A NACA mode is also available, but functionality is limited.

Two multi objective optimization (MOO) algorithms are available – NSGA2 and CMOPSO.

User inputs are required in ***config.py***, and the script is executed with ***airfoilOpt.py***. Once the optimization is complete, the history and results can be visualized with ***postProc_Multi.ipynb***

Tested and developed on Ubuntu 22.04

Methodology

The airfoil is parameterized with either the 4 digit NACA and 11 coefficient PARSEC format. Main functionality is implemented for the PARSEC parametrization. The user specifies the design space by providing a range for each coefficient in the parametrization format. The user can also provide a seed airfoil to bias initial design space exploration

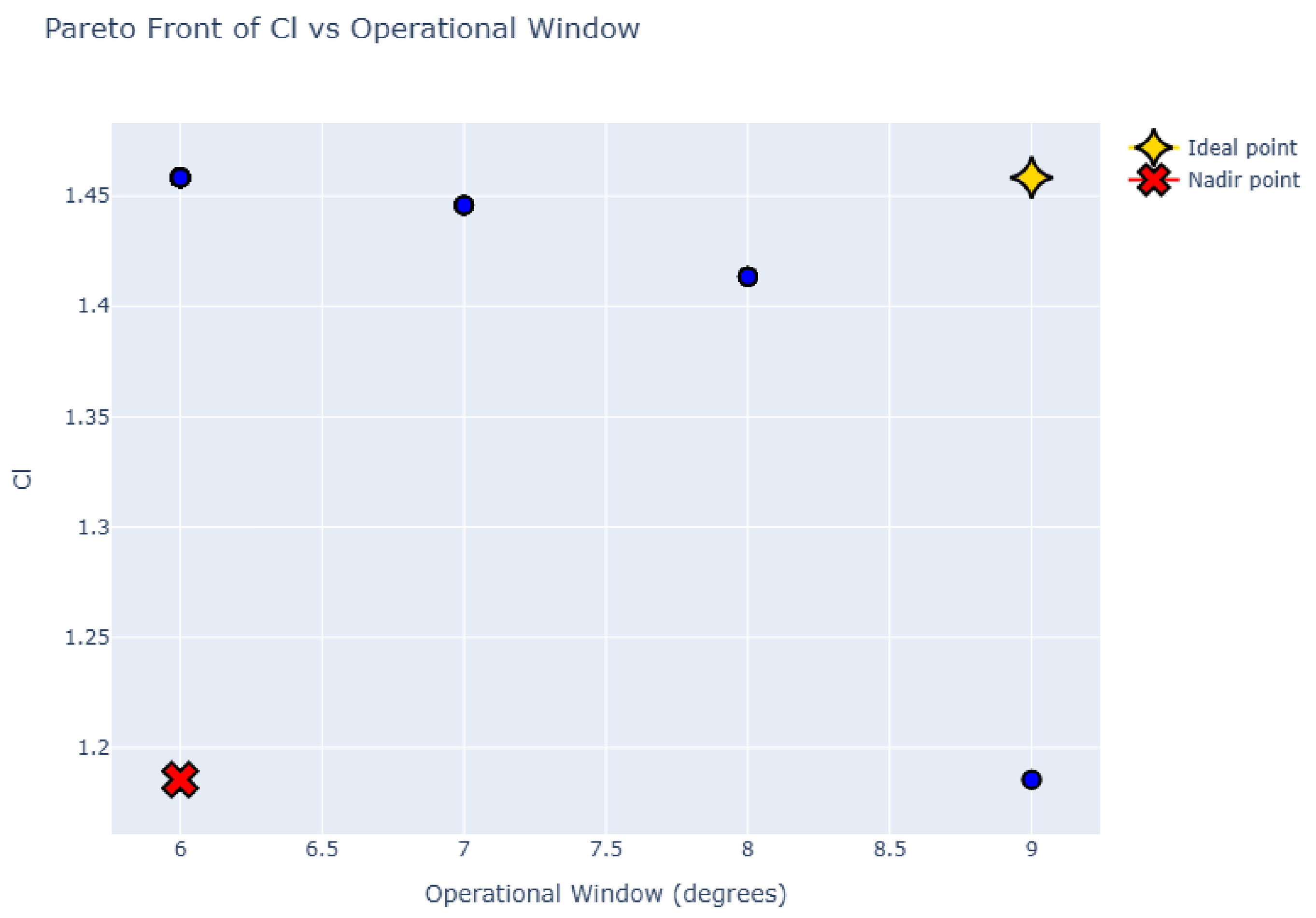
The python library ***pymoo*** is used to implement the MOO algorithms and define the optimization problem.

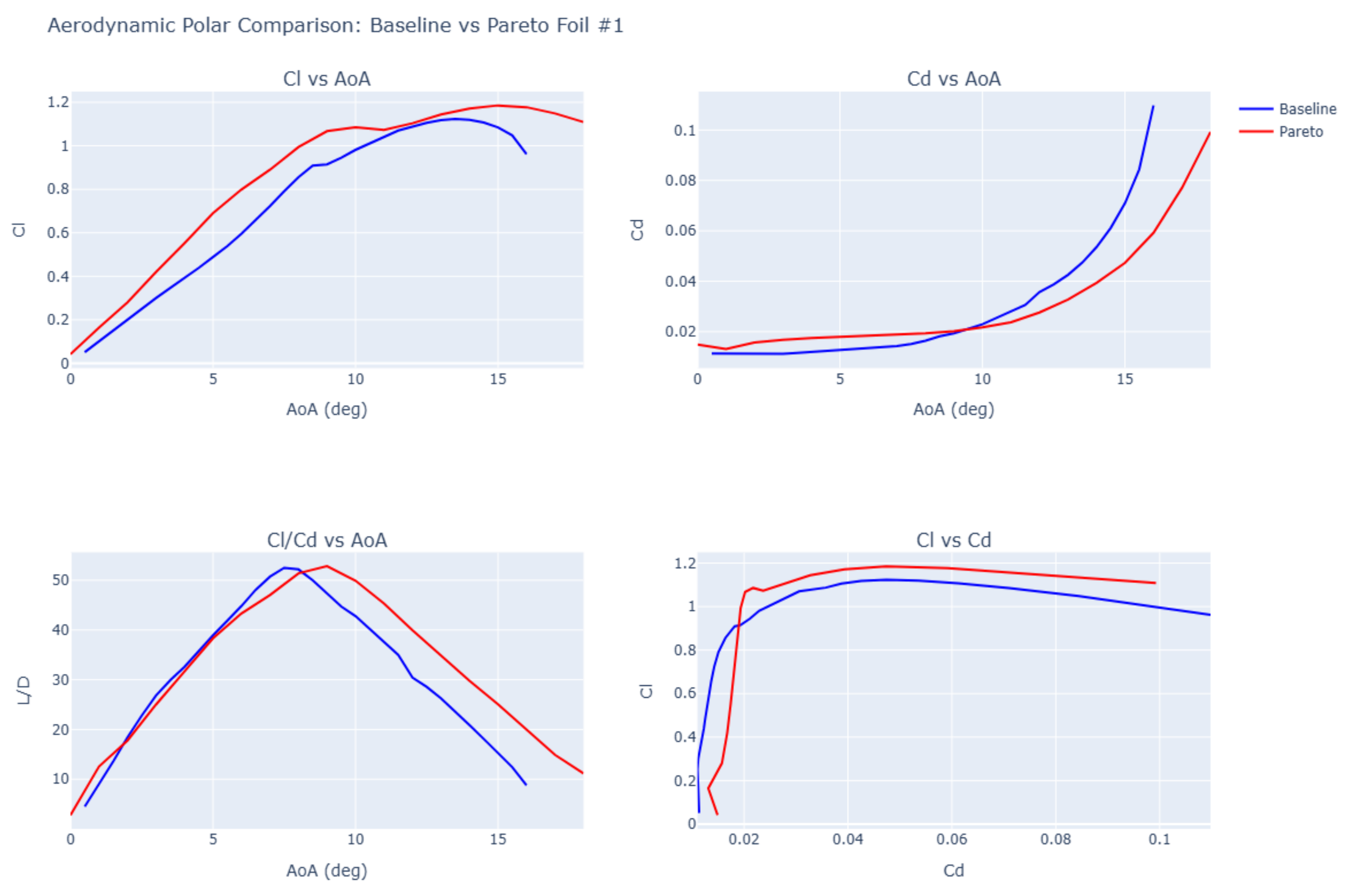
The objective function is to maximize $C_{l_{max}}$ and operational window measured in degrees. Both values are normalized with respect to a user specified '*baseline*'.

During an optimization run, the algorithm generates an airfoil in the design space, evaluates the airfoil in XFOIL and the objective function is evaluated. Any errors during evaluation are penalized during scoring. Once the run is complete, a pickle file is written out containing the run history.

Results

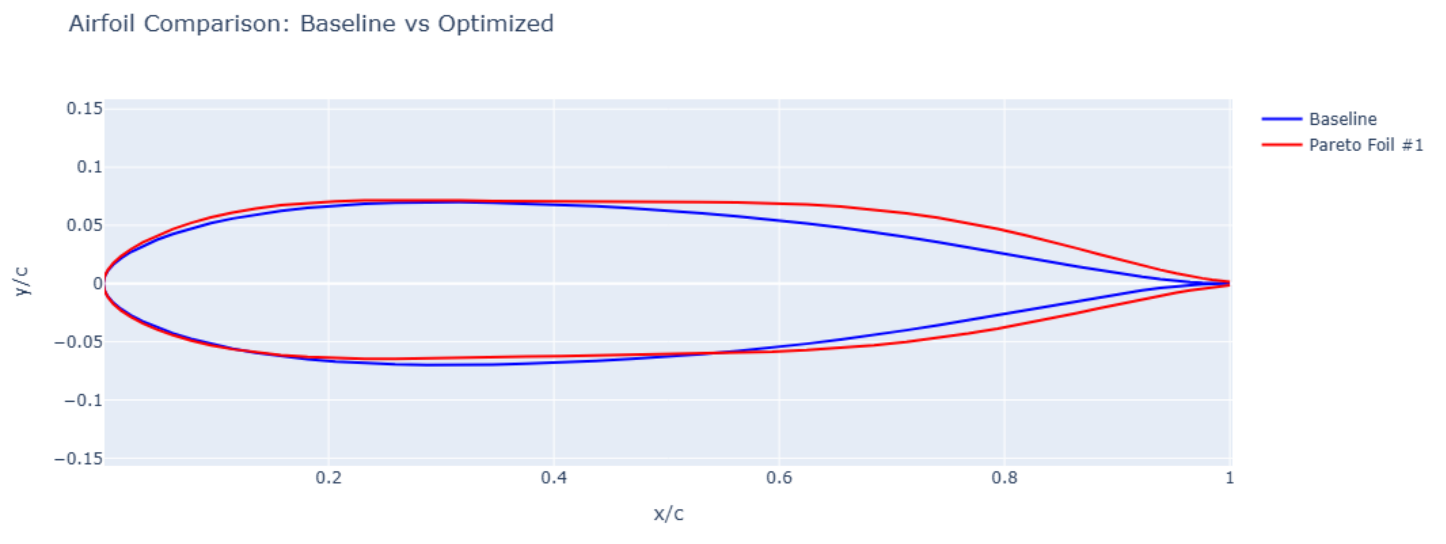
Using NACA0012 as a seed, the optimizer was able to produce designs that theoretically improve upon the baseline, with $C_{l_{max}}$ **improvements from 5% to 29%** ,and **operational window improvements from 100% to 200%**





Results (contd.)

The below image shows the airfoil for the second point on the pareto front from the left. The previous polar comparison uses this airfoil and the baseline NACA0012 airfoil



Limitations

- Results depend greatly on chosen bounds for PARSEC parameters
- PARSEC parametrization can produce airfoils that are not smooth (ex – The airfoil highlighted above)

Improvements

- Geometry smoothness and feasibility checks could be implemented
- A different parametrization method like CSV could also remediate the aforementioned limitations